

Cost & Dev-Hours Levers — Full Launch (Load Test + PWA + Native Apps)

Goal: land a **full launch** — web PWA, Android, iOS, plus a load test — while keeping **development hours** and **runtime cost** at the low end, and keeping the **new AI/catalog features from adding to the bill**. This is the complete list (everything we've used before + new levers created by the recent AI/serverless-GPU/marketplace work). Nothing here requires new code — every lever is a config flip, a prebuilt toggle, or an owner action.

Bottom line up front: the **dev-hours estimate stays flat** at roughly **30-45 h (~\$2,250-\$3,375)** for the full PWA + Android + iOS launch, because every new feature was built env-gated and off-by-default — so it adds **\$0 to the launch estimate**, exactly like the Claude switch and scale toggles did. The **new runtime cost** (AI text + images) is capped and can launch at **near \$0**, then scale with usage.

1. Dev-hours levers (one-time build/launch) — reuse what we did before

1. **Railway single-service mode** — one deploy runs backend + Postgres + scheduler + the built SPA (AUTO_MIGRATE=1, SCHEDULER_INLINE=1, FRONTEND_DIR=./dist). No separate infra wiring.
2. **Capacitor wrapper, one codebase** — the PWA build is the app; Android and iOS are thin wrappers. No native rewrite, no per-platform code.
3. **Cloud iOS CI (no Mac)** — Codemagic / GitHub Actions build the iOS app in the cloud. No Mac purchase, no separate iOS engineer.
4. **Owner does the signups** — API keys and account creation (API-KEYS-WORKSHEET.md) are owner tasks, not billable dev hours.
5. **validate.sh before any cloud work** — proves the build is green first; kills the #1 cause of high-end blowout (discovery + rework in the cloud).
6. **Prebuilt scale toggles, dormant** — Redis / read-replica / SQS are one env var each and no-ops until set. Scaling later is a flip, never a rewrite.
7. **Claude switch + image toggle** — LLM_PROVIDER, IMAGE_PROVIDER are env flips; swapping the AI brain or image engine is zero code change.
8. **Everything env-gated with safe-OFF defaults** — every new feature (catalog images, affiliate feeds, browse-node expansion, card charging, cash-out, SMS) degrades gracefully when unconfigured, so launch never blocks on a provider and nothing needs reworking to add later.
9. **Reuse the execution kit verbatim** — deploy-kit/ ships the exact Railway JSON, Dockerfile, CI YAML, and checklists, so the developer runs near the *low* end of every phase.

2. Runtime-cost levers for the NEW AI / catalog features (keep monthly cost near what it was)

1. **Serverless GPU images (AWS Bedrock or scale-to-zero)**

- SageMaker**) — pay per image (fractions of a cent) or per-second of generation; **no idle cost, no per-image API list price.**
Cheapest path at catalog scale.
2. **Template-once images** — a product image is generated **one time** and reused across all 88 country catalogs (only the flag/price change). That's ~88× fewer images than generating per country.
 3. **CATALOG_IMAGES_ENABLED kill switch + CATALOG_IMAGES_MAX_PER_RUN cap** — bound image spend per run, or launch **text-only at \$0 image cost** and backfill images after launch.
 4. **AI_DAILY_SPEND_CAP_USD** — a hard global brake: LLM calls are refused once the day's estimated spend crosses the cap. Set it to your comfort number and the AI literally cannot overspend.
 5. **Small-tier models for cheap work** — catalog copy, browse-node names, and translations use the small model tier (LLM_MODEL_SMALL / CLAUDE_MODEL_SMALL — gpt-4o-mini / Claude Haiku), a fraction of the large-model price.
 6. **Paced background jobs** — the catalog seed, category images, and browse-node expansion run in bounded per-run batches on a schedule, so cost is spread smoothly instead of a spike (the whole ~905-call browse-node expansion is only a few dollars total, paced over days).
 7. **Translation caching** — each string is translated once per language and reused; repeat views cost nothing.
 8. **Free-tier external calls, cached** — exchange rates (exchangerate-api, cached 1 h) and geo-IP country (ipapi.co, cached 24 h) are free at launch volume; both have static fallbacks so they can run at **zero external calls** if you prefer.
 9. **Everything falls back to a free/degraded mode** — no LLM key → template catalog listings; no image provider → text-only listings; no affiliate creds → neutral search; no live FX → static table. You can launch at **\$0 AI spend** and turn providers on when you're ready.
 10. **Affiliate + original-catalog model = \$0 inventory** — no stock to buy, no fulfillment to run; original listings + affiliate links earn commission instead of costing inventory.
 11. **Single-instance launch posture** — Railway ~\$10-30/mo at launch traffic; all scale knobs off until the load test proves you need one.

3. Load-test levers

1. **Defer the full AWS load test** until real traffic justifies it — the plan (LOAD-TEST-PLAN.md) is ready; you don't pay for a fleet before you have users.
2. **Cheap first pass** — run k6 / Artillery from one small box (or free tier) against staging to find the first ceiling, instead of standing up an AWS load generator fleet.
3. **Spin up, test, tear down** — when you do the AWS load test, use spot instances and destroy them the same day; it's an hours-long spend, not a standing cost.
4. **The scale toggles make the load test decisive** — it tells you the **one** knob to flip (Redis, replica, or SQS), so you never speculatively over-provision.

4. New additions to the Execution Kit (add these files/steps)

1. **AI-COST-WORKSHEET** — owner sets, before launch: AI_DAILY_SPEND_CAP_USD, CATALOG_IMAGES_MAX_PER_RUN, CATALOG_COUNTRIES (start with 1), IMAGE_PROVIDER, and the small/large model ids. One page, no dev time.
2. **"Text-only launch, images later" path** — launch with CATALOG_IMAGES_ENABLED=0 (\$0 images), flip it on post-launch to

- backfill. Documented as a checklist step.
3. **“One-country launch” posture** — launch CATALOG_COUNTRIES=US only; adding a country later is a clone of existing templates (no new images, no new dev).
 4. **Provider cost cheat-sheet** — Bedrock Nova Canvas $\approx$ \$0.04/img, Titan Image $\approx$ \$0.01/img, SDXL on scale-to-zero SageMaker $\approx$ fractions of a cent; small-tier LLM $\approx$ \$0.15-\$0.60 per 1M tokens. Lets the owner predict the monthly number before turning anything on.
 5. **Catalog ramp plan** — start CATALOG_LISTINGS_PER_COUNTRY small (e.g. 80), grow after launch, so the initial seed cost is tiny and breadth is still instant.

5. Other suggestions

1. **Launch the catalog small** — one product per department (~40 items) gives instant Amazon-style breadth for a few dollars of AI spend; grow depth afterward.
 2. **Turn browse-node expansion OFF at launch** — the 905 subcategories already exceed most needs; the ~21,700 browse nodes are a post-launch growth feature, not a launch blocker.
 3. **Free infra tiers** — Cloudflare (DNS/CDN), Sentry free tier (errors), Railway trial credit.
 4. **Keep card charging / cash-out OFF at launch** — the closed-loop points economy launches with no payment-processor integration cost or compliance blocker; enable when the processor + legal are set.
 5. **Keep the iOS review clean** — demo login + “merit, not gambling” framing avoids rejection rounds, the single biggest budget risk on the native side.
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The number, restated

- **Dev hours:** ~30-45 h (~\$2,250-\$3,375) for full PWA + Android + iOS — **unchanged**, because the new features are prebuilt and off-by-default.
- **Runtime at launch:** hosting ~\$10-30/mo; AI/images **capped and optional** — realistically **\$0-100 in month one** depending on catalog size and whether images are on, and it can never exceed your AI_DAILY_SPEND_CAP_USD.
- **Load test:** an hours-long spot-instance spend, deferred until traffic warrants it.